



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 • STANDARD 6.GR.4

# Determine Surface Area of Prisms

Lesson 5-7 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 3 Enrichment



### TODAY'S OBJECTIVES

**Content Objective:** I can find the surface area of rectangular and triangular prisms.

**Language Objective:** I can explain my work using the words surface area, rectangular prism, triangular prism, and face.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Surface area

Rectangular prism

Triangular prism

Face

Net

Lateral face



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The total area of all the faces of a solid is its .

2

A solid with six rectangular faces is a .

3

A solid with two triangular faces and three rectangular faces is a

.

4

A flat surface of a solid figure is a .

5

A flat pattern that folds into a solid is a .

6

A side face of a prism that is not a base is a .

## Write About the Math

**How much surface area needs staining, and how many cans of stain are needed?**

*¿Cuánta área de superficie hay que teñir, y cuántas latas de tinte se necesitan?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Surface area**  
*Área de superficie*

☐ **Rectangular prism**  
*Prisma rectangular*

☐ **Triangular prism**  
*Prisma triangular*

☐ **Face**  
*Cara*

☐ **Net**  
*Plantilla (desarrollo plano)*

**Model:** For both, you add the areas of all the faces, but the shapes and number of faces differ. — Students explain both methods add up all face areas, but a rectangular prism has 6 rectangular faces while a triangular prism has 5 faces (2 triangles + 3 rectangles).

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The chest's SA is \_\_\_\_ ft<sup>2</sup> because  $SA = 2(\text{____} \times \text{____}) + 2(\text{____} \times \text{____}) + 2(\text{____} \times \text{____}) = \text{____} + \text{____} + \text{____} = \text{____}$ . You need \_\_\_\_ can(s) because  $\text{____} \div 50 = \text{____}$ .**
- **My answer is \_\_\_\_ because \_\_\_\_.**  
*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My evidence is \_\_\_\_, so \_\_\_\_.**  
*Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

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### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Triangular prism
4. **Notes 4:** Face
5. **Notes 5:** Net
6. **Notes 6:** Lateral face
7. **Watch for this mistake:** *A common mistake in Surface Area of Prisms is confusing surface area with volume — multiplying  $l \times w \times h$  (which finds the space INSIDE the prism, in cubic units) instead of adding the areas of all the faces ( $SA = 2lw + 2lh + 2wh$ , in square units). A second common slip is forgetting to double one face pair, like writing  $2(l \times w) + 2(l \times h) + (w \times h)$  instead of  $+ 2(w \times h)$  — always count all 6 faces (or all 5 for a triangular prism) before you submit.*
8. **Try It 1:** C.  $288 \text{ in}^2$  — *Two triangle ends:  $2 \times (\frac{1}{2} \times 8 \times 6) = 48 \text{ in}^2$ . Three rectangles:  $(8 + 6 + 10) \times 10 = 24 \times 10 = 240 \text{ in}^2$ .  $SA = 48 + 240 = 288 \text{ in}^2$ .*
9. **Try It 2:** — *A net shows every face laid out flat, so folding it up proves that surface area is just the sum of the areas of all the shapes on the net.*